



Bayesian modeling of extreme droughts risk

Presented by

Safoorah Sabir

Quaid-e-Azam University, Islamabad, Pakistan

Co Authors

Dr. Touqeer Ahmad

CREST, ENSAI, France

Shafique Ur Rehman

International College, University of Chinese Academy of
Sciences, Beijing, China

November 14, 2024



Outline

◀ Motivation

- Drought
- Types of Drought
- Motivation of the study

◀ Methodology

- Study Area
- Extreme Value Theory
- Selection of Informative Priors
- Sen's Slope Test
- Stationary and Non-Stationary SPI's
- Classical Modeling
- Bayesian Modeling
- Risk Assessment

◀ Results

◀ Conclusion and Future Work



What is Drought?

Definitions

- ◀ Drought is a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is a slow-onset disaster characterized by the lack of precipitation, resulting in a water shortage. (WHO)
- ◀ A drought is a period when an area or region experiences below-normal precipitation. The lack of adequate precipitation, either rain or snow, can cause reduced soil moisture or groundwater, diminished stream flow, crop damage, and a general water shortage. (National Geographic)



Countined...

Drought

- ◀ **DURATION:** A drought can last for months or years or may be declared in the early days if low rain perception is clear for some period.
- ◀ **IMPACTS:** Droughts can have a substantial impact on the ecosystem and agriculture of the affected region and harm the local population and economy.



Types of Drought



4 TYPES OF DROUGHT

AGRICULTURAL DROUGHT

Agricultural Droughts occur when there is not enough moisture in the soil to sustain the growth of crops.

SOCIOECONOMIC DROUGHT

Socioeconomic Droughts occur when the water supply is too low to support human and environmental needs

HYDROLOGICAL DROUGHT

Hydrological Droughts occur when there is a lack of surface and subsurface water supply.

METEOROLOGICAL DROUGHT

Meteorological Droughts are region-specific; they occur when an area receives less rainfall than it normally should.

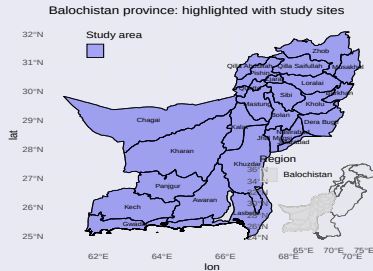


Motivation of the Study

- ◀ Safeguarding water and food security in Baluchistan.
- ◀ Protecting livelihoods from climate change impacts.
- ◀ Understanding and mitigating drought phenomena.
- ◀ Empowering decision-makers with knowledge and tools.
- ◀ Ensuring well-being of present and future generations.

Methodology

Study Area



- ▶ The series of each is divided into two 20-year duration subseries: Period I (1984–2003) and Period II (2004–2023).



Extreme Value Theory

- ◀ Extreme value theory deals with rare events and comprises through block maxima and peak over threshold.
- ◀ Extreme values occurred at the tails of a probability distribution.
- ◀ Models: GEV and GPD.



Informative Priors

◀ Reg (Afghanistan)

- Distance $\sim$ 100 km (Northern Baluchistan)
- Dry, mountainous region
- Surrounded by mountains that block monsoon winds
- Exhibits climatic trends parallel to northern Baluchistan

◀ Zahedan (Iran)

- Distance $\sim$ 160 km (Western Baluchistan)
- Arid climate with low precipitation
- High temperatures typical of the region
- Geographic and climatic patterns similar to western Baluchistan

◀ Dadu (Sindh)

- Distance $\sim$ 180 km (Eastern Baluchistan)
- Arid, desert-edge climate
- Close proximity to the Indus River
- Reflects the arid conditions prevalent in eastern Baluchistan



Sen's Slope Test

◀ Hypothesis

H_0 : there is no monotonic trend existing.

H_1 : there is a monotonic trend exists.

◀ Test Statistics

The Sen's Slope (δ) is defined as

$$\delta = \text{median} \left\{ \frac{y_j - y_k}{j - k} : k < j \right\}, \quad (1)$$

Stationary and Non-Stationary Standardized Precipitation Index (SPI & NSSPI)

- ▶ SPIs or NSSPIs series are constructed for each period of all stations.
- ▶ Calculation based on the six-month sum of precipitation data for the corresponding period.
- ▶ **NSSPI** is constructed when a monotonic trend is detected in the data; otherwise, **SPI** is used if no trend is found.

Table 1: Classification of drought levels based on SPI or NSSPI values

SPI/NSSPI values	Drought category
≥ 2.00	Extremely humid
$1.50 \leq \text{SPI} \leq 2.00$	Very humid
$1.00 \leq \text{SPI} \leq 1.50$	Humid
$-1.00 \leq \text{SPI} \leq 1.00$	Normal
$-1.50 \leq \text{SPI} \leq -1.00$	Dry
$-2.00 \leq \text{SPI} \leq -1.50$	Very dry
$\text{SPI} \leq -2.00$	Extremely dry

Source: Labedzki, L. (2006). Susze rolnicze. Zarys problematyki oraz metody monitorowania i klasyfikacji. Woda. Srodowisko. Obszary Wiejskie. Rozprawy Naukowe i Monografie, (17):1–107.



Classical Modeling

Generalized Extreme Value (GEV) distribution

- ▶ GEV distribution models the block maxima.
- ▶ The inverse CDF of GEV is written as

$$G_{\alpha, \beta, \kappa}(x) = \begin{cases} \exp \left\{ - \left[1 + \kappa \left(\frac{x - \alpha}{\beta} \right) \right]^{-\frac{1}{\kappa}} \right\} & \text{if } \kappa \neq 0, \kappa \left(\frac{x - \alpha}{\beta} \right) > 0 \\ \exp \left\{ \exp \left(\frac{x - \alpha}{\beta} \right) \right\} & \text{if } \kappa = 0, x \in \mathbb{R}. \end{cases} \quad (2)$$

- ▶ The log-likelihood function for unknown parameters vector $\theta = (\alpha, \beta, \kappa)$ is defined as

$$l(\theta) = -n \log \beta - \left(1 + \frac{1}{\kappa} \right) \sum_{i=1}^n \log \left[1 + \kappa \frac{(x_i - \alpha)}{\beta} \right] - \sum_{i=1}^n \left[1 + \kappa \frac{(x_i - \alpha)}{\beta} \right]^{-\frac{1}{\kappa}} \quad (3)$$

Bayesian Modeling

Bayesian Modeling for Non-Informative Priors

- ▶ The GEV distribution parameters $\theta = (\alpha, \beta, \kappa)$ are considered random variables for which we identify the prior distributions.
- ▶ θ in terms of posterior PDF, is given by

$$\pi(\theta|x) = \frac{L(x|\theta)g_{\theta}(\theta)}{\int_{\Theta} L(x|\theta)g_{\theta}(\theta)d\theta}, \quad (4)$$

- ▶ The joint distribution for θ was supposed to be written as

$$g_{\theta}(\theta) = g_{\theta}(\alpha, \beta^* = \log(\beta), \kappa) = g_{\alpha}(\alpha)g_{\beta^*}(\beta^*)g_{\kappa}(\kappa). \quad (5)$$

- ▶ For marginals, we consider non-informative priors as

$$g_{\alpha}(\alpha) \sim \mathcal{N}(0, 10000) \quad g_{\beta^*}(\beta^*) \sim \mathcal{N}(0, 10000) \quad g_{\kappa}(\kappa) \sim \mathcal{N}(0, 100). \quad (6)$$

Construction of Informative Priors

- ◀ The quantile difference for informative priors is:

$$\tilde{x}_{p_i} = x_{p_i} - x_{p_{i-1}}, \quad i = 1, 2, 3 \quad (7)$$

- ◀ The marginal distributions are written as

$$\tilde{x}_{p_i} \sim \mathcal{G}(\gamma_i, \delta_i), (\gamma_i, \delta_i) > 0, \quad i = 1, 2, 3 \quad (8)$$

- ◀ The joint distribution can be written as

$$g(x_{p_1}, x_{p_2}, x_{p_3}) \propto \tilde{x}_{p_1}^{\gamma_1-1} e^{-\delta_1 \tilde{x}_{p_1}} \prod_{i=2}^3 \tilde{x}_{p_i}^{\gamma_i-1} e^{-\delta_i \tilde{x}_{p_i}}. \quad (9)$$



Risk Assessment

- ◀ Drought risk is the likelihood of experiencing a drought.
- ◀ The risk is defined as

$$R_{\mathcal{D}}(G_{\alpha,\beta,\kappa}, d_{cr}) = \mathcal{D} \quad (10)$$

and

$$R_{\mathcal{D}_Q}(G_{\alpha,\beta,\kappa}^{-1}, p_{cr}) = \mathcal{D}_Q, \quad (11)$$

where $G_{\alpha,\beta,\kappa}^{-1}$ is inverse CDF of GEV distribution.



Estimated Trend

Table 2: Estimated time trend and its significance testing for Period I and II in each region. P-values in bold show the trend existence.

Site	Period	Sen's slop	Test Statistic	P-value
Gawadar	I	0.0066	3.4018	0.0007
	II	0.0013	0.5816	0.5609
Musakhel	I	-0.0097	-1.3218	0.1862
	II	0.0128	1.4820	0.1384
Ziarat	I	-0.0162	-2.8885	0.0039
	II	0.0188	2.2828	0.0224



Estimated Parameters

Table 3: Estimator values for parameters of distribution functions $G_{\mu,\sigma,\kappa}$

Station	Period	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\kappa}$
Gawadar	I	-0.433 (0.064)	0.871 (0.046)	-0.111 (0.051)
	II	-0.382 (0.070)	0.949 (0.050)	-0.223 (0.050)
Musakhel	I	-0.415 (0.065)	0.900 (0.045)	-0.137 (0.041)
	II	-0.448 (0.058)	0.801 (0.040)	-0.033 (0.039)
Ziarat	I	-0.293 (0.075)	1.049 (0.055)	-0.403 (0.039)
	II	-0.407 (0.066)	0.913 (0.045)	-0.159 (0.042)

Estimated Parameters

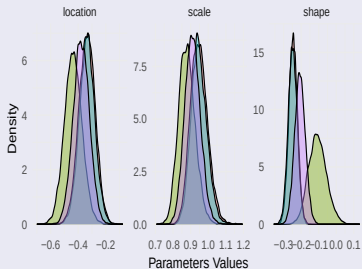
Table 4: Posterior location, scale, and shape for GEV parameters of different weather stations via noninformative and informative priors

Stations	Periods	NonInformative prior			Informative prior (Reg)			Informative prior (Zabedan)			Informative prior (Dadu)		
		$\hat{\alpha}$	$\hat{\beta}$	$\hat{\kappa}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\kappa}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\kappa}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\kappa}$
Gawadar	I	-0.433 (0.066)	0.880 (0.047)	-0.108 (0.051)	-0.340 (0.058)	0.945 (0.046)	-0.249 (0.026)	-0.378 (0.058)	0.913 (0.045)	-0.207 (0.031)	-0.332 (0.057)	0.952 (0.046)	-0.245 (0.024)
	II	-0.385 (0.071)	0.957 (0.051)	-0.220 (0.050)	-0.355 (0.061)	0.973 (0.047)	-0.262 (0.029)	-0.283 (0.062)	1.032 (0.051)	-0.348 (0.027)	-0.406 (0.061)	0.940 (0.044)	-0.209 (0.035)
Musakhel	I	-0.418 (0.066)	0.907 (0.046)	-0.131 (0.042)	-0.345 (0.057)	0.941 (0.045)	-0.230 (0.021)	-0.381 (0.058)	0.921 (0.043)	-0.200 (0.026)	-0.334 (0.057)	0.955 (0.045)	-0.234 (0.020)
	II	-0.451 (0.058)	0.808 (0.041)	-0.028 (0.039)	-0.367 (0.052)	0.842 (0.039)	-0.145 (0.020)	-0.31 (0.050)	0.875 (0.039)	-0.189 (0.016)	-0.426 (0.052)	0.807 (0.036)	-0.087 (0.025)
Ziarat	I	-0.301 (0.075)	1.055 (0.056)	-0.393 (0.040)	-0.302 (0.066)	1.053 (0.050)	-0.389 (0.030)	-0.316 (0.069)	1.043 (0.051)	-0.378 (0.035)	-0.326 (0.064)	1.040 (0.048)	-0.344 (0.028)
	II	-0.409 (0.066)	0.921 (0.046)	-0.156 (0.043)	-0.360 (0.058)	0.945 (0.044)	-0.229 (0.026)	-0.290 (0.058)	0.999 (0.049)	-0.301 (0.023)	-0.414 (0.059)	0.911 (0.041)	-0.174 (0.030)

Posterior Densities

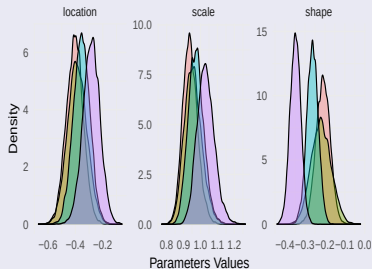
**Posterior Densities of GEV Parameters
Gawadar Period I**

Priors ■ Dadu ■ Noninformative ■ Reg ■ Zahedan



**Posterior Densities of GEV Parameters
Gawadar Period II**

Priors ■ Dadu ■ Noninformative ■ Reg ■ Zahedan

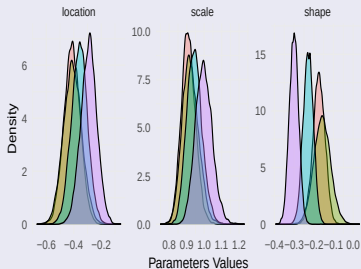


Posterior Densities of Gawadar

Posterior Densities

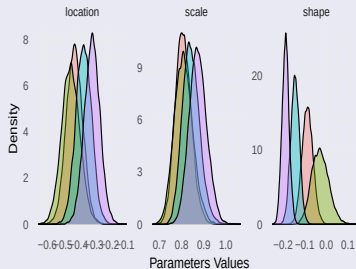
Posterior Densities of GEV Parameters
 Musakhel Period I

Priors ■ Dadu ■ Noninformative ■ Reg ■ Zahedan



Posterior Densities of GEV Parameters
 Musakhel Period II

Priors ■ Dadu ■ Noninformative ■ Reg ■ Zahedan

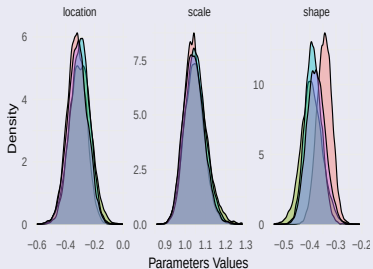


Posterior Densities of Musakhel

Posterior Densities

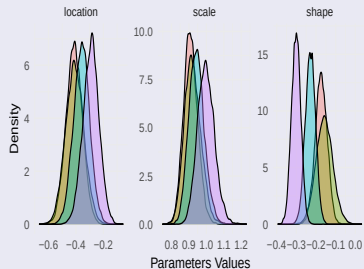
Posterior Densities of GEV Parameters
 Ziarat Period I

Priors ■ Dadu ■ Noninformative ■ Reg ■ Zahedan



Posterior Densities of GEV Parameters
 Ziarat Period II

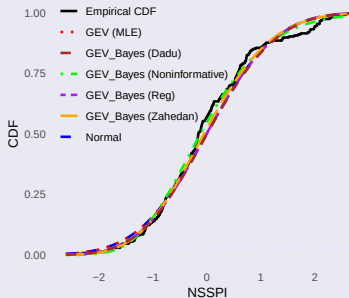
Priors ■ Dadu ■ Noninformative ■ Reg ■ Zahedan



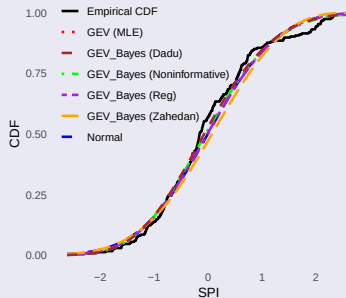
Posterior Densities of Ziarat

CDF Plots for Gawadar

CDF plot (Gawadar Period I)

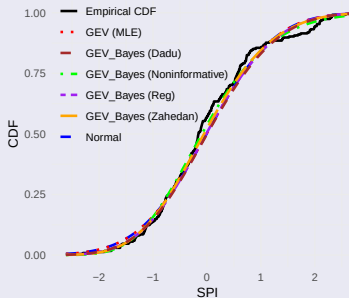


CDF plot (Gawadar Period II)

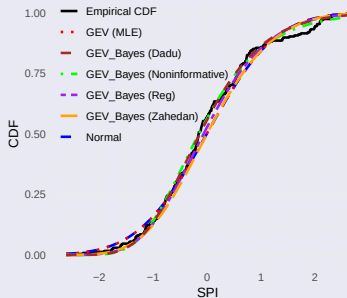


CDF Plots for Musakhel

CDF plot (Musakhel Period I)

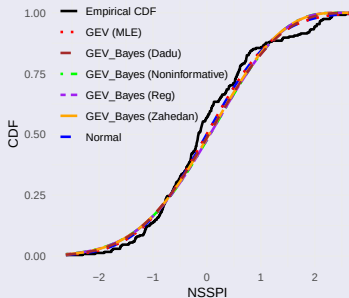


CDF plot (Musakhel Period II)

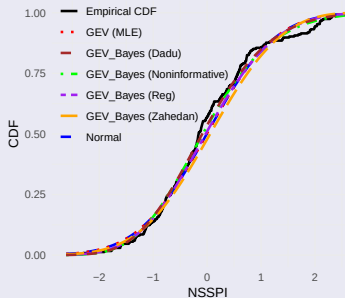


CDF Plots for Ziarat

CDF plot (Ziarat Period I)

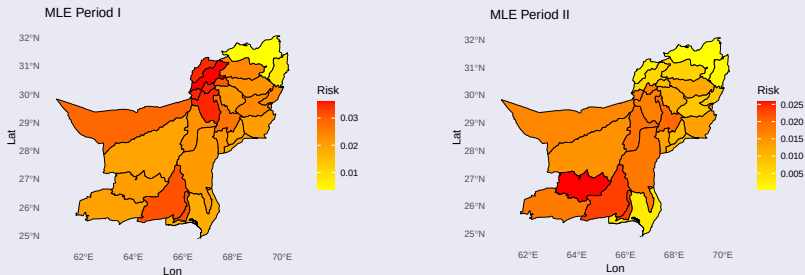


CDF plot (Ziarat Period II)



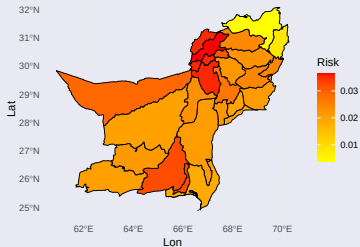
Drought Risk Measures

Drought Risk for MLE

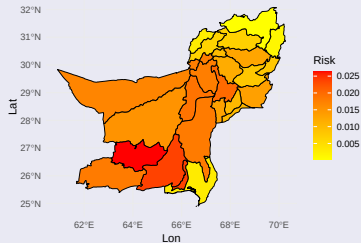


Drought Risk for Non Informative Prior

NIP Period I

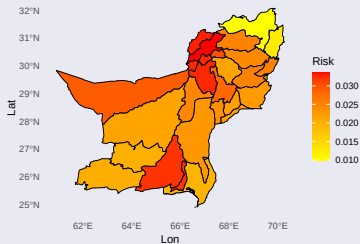


NIP Period II

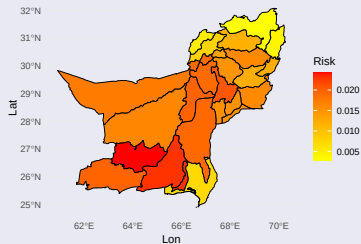


Drought Risk for Reg (Afgahnistan)

Reg Period I

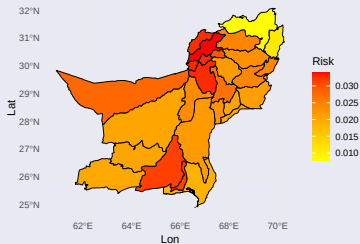


Reg Period II

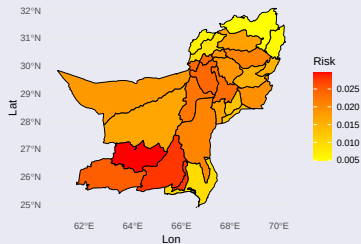


Drought Risk for Zahedan (Iran)

Zahedan Period I

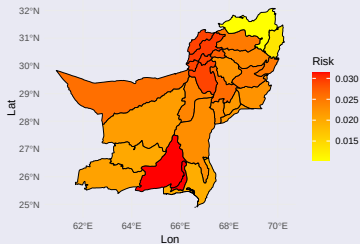


Zahedan Period II

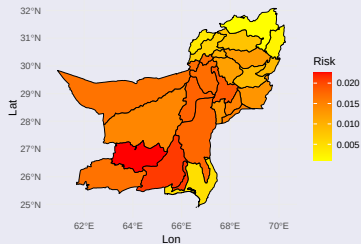


Drought Risk for Dadu (Sindh)

Dadu Period I



Dadu Period II





Conclusion

Conclusion

- ◀ Despite extreme observations in the data, the proposed framework is flexible and efficient for measuring drought.
- ◀ There's no drought risk in north part of the Baluchistan for all techniques.
- ◀ The results declare that north is safest area for living as compare to south.
- ◀ Lasbela district and North-western side of the province is also safe in Period II.
- ◀ These results shows that the climate change/global warming has changed the province's drought risk (high to low) conditions in last 20 years.



Future Work

Work in progress

- ◀ This framework can be extended for NSSPI by considering the effect of other environmental covariates.
- ◀ GEV can be replaced by GPD if one is interested in modeling the threshold exceedances.



THANK YOU

Any Questions?